

WMF SERIES (20 TO 2,000 CFM)

High-Efficiency In-Line Compressed Air Filters

Four-Stage Progressive Coalescing & Particulate Filtration Delivering ISO 8573-1 Class 1 Air Purity Down to 0.01 Micron & 0.003 ppm Oil Content



0.01 MICRON

Grade Y Coalescing
Micro-Filtration

0.003 PPM

Grade A Activated
Carbon Vapor Removal

ISO 8573-1

Compliant with
International Air Purity
Standards

16 BAR RATED

Corrosion-Proof Die-Cast
Aluminum Housings

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WMF Series in-line compressed air filters remove solid particulates, liquid water aerosols, and oil mist from industrial air networks. Available in four progressive filtration grades—Grade P (pre-filter), Grade X (general purpose), Grade Y (micro-coalescer), and Grade A (activated carbon vapor absorber)—WMF filters ensure compliance with ISO 8573-1 air purity standards, protecting expensive automated machinery, instrumentation, and food packaging from contamination.

Working Principle: Compressed air enters the die-cast aluminum housing where internal aerodynamic flow channels guide air through multi-layered borosilicate microfiber filtration sleeves. Large particulates are captured via direct interception; sub-micron oil droplets collide via Brownian motion and coalesce into heavy liquid drops that drain into the quiet bottom sump zone for automatic expulsion. In Grade A filters, an activated carbon deep-bed adsorbs hydrocarbon odors and vapors, delivering clinically pure, breathable air.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Raw Air → Grade P (3μ Pre-filter) → Air Dryer → Grade X (1μ Particulate) → Grade Y (0.01μ Coalescer) → Grade A (0.003 ppm Vapor) → Ultra-Clean Sterile Process Air

Four-Stage Progressive Filtration Architecture

Matches filtration precision exactly to process requirements: Grade P (3μm / 5 ppm oil), Grade X (1μm / 0.5 ppm), Grade Y (0.01μm / 0.01 ppm), and Grade A (0.003 ppm oil vapor & odor).

Alocrom Passivated Die-Cast Aluminum Housings

Housings are pressure die-cast from high-grade aluminum, treated with Alocrom anti-corrosion chemical passivation, and electrostatically powder coated. Guaranteed 10-year housing life against pinhole leaks.

Integrated Pop-Up Differential Pressure Indicator

Equipped with a top-mounted differential pressure gauge or visual pop-up indicator that signals when the element is saturated (>0.35 bar ΔP), preventing unnecessary energy waste.

Automatic Condensate Evacuation Standard

Every housing includes an internal float-type automatic drain or external zero-air-loss electronic drain that continuously vents coalesced liquid oil and water without human intervention.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Max Working Pressure: 16.0 bar g | Max Working Temp: 65°C (Grade A: 30°C) |
Housing Material: Alocrom Treated Die-Cast Aluminum | Element Media: Hydrophobic Borosilicate Microfiber

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	PIPE CONN	GRADE OPTIONS	DIMENSIONS (L×W×H MM)	WEIGHT KG
WMF 004	20 CFM	34	1/2" BSP	P, X, Y, A	90 × 90 × 220	1.1 kg
WMF 008	45 CFM	76	1/2" BSP	P, X, Y, A	90 × 90 × 260	1.3 kg
WMF 015	85 CFM	144	3/4" BSP	P, X, Y, A	110 × 110 × 300	1.8 kg
WMF 025	140 CFM	238	1" BSP	P, X, Y, A	110 × 110 × 380	2.2 kg
WMF 035	200 CFM	340	1.25" BSP	P, X, Y, A	140 × 140 × 440	3.6 kg
WMF 060	350 CFM	595	1.5" BSP	P, X, Y, A	140 × 140 × 520	4.2 kg
WMF 100	600 CFM	1,020	2" BSP	P, X, Y, A	170 × 170 × 630	6.8 kg
WMF 150	850 CFM	1,445	2.5" BSP	P, X, Y, A	170 × 170 × 750	8.5 kg
WMF 250	1,400 CFM	2,380	3" Flange	P, X, Y, A	250 × 250 × 850	18.0 kg
WMF 350	2,000 CFM	3,400	3" Flange	P, X, Y, A	250 × 250 × 980	24.0 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- Grade P: Pre-filtration (3 micron, removes dust and liquid water drops).
- Grade X: General purpose particulate removal (1 micron, oil aerosol <0.5 mg/m³).
- Grade Y: High efficiency coalescing (0.01 micron, oil aerosol <0.01 mg/m³).
- Grade A: Activated carbon vapor adsorption (oil vapor & hydrocarbon odor <0.003 mg/m³).

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

Pharmaceutical Cleanrooms & Pill Packaging

Guarantees zero oil contamination in sterile air lines compliant with FDA Class 1 compressed air standards.

Food, Beverage & Dairy Processing

Eliminates foul hydrocarbon odors and oil taste from compressed air contacting food containers and dairy bottles.

Automated Spray Painting & Powder Coating

Stops micro-droplets of oil from creating craters, blisters, and surface blemishes on vehicle bodies.

Laser Cutting Assist Gas Pre-Treatment

Prevents optical mirror fogging and cutting lens burn-through caused by hydrocarbon vapor contamination.

Precision Coordinate Measuring Machines (CMM)

Protects frictionless air bearing pads from microscopic dust particle scoring and premature mechanical wear.

Engineering Comparison: Win WMF vs. Conventional Pipeline Strainers		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Filter Media	Hydrophobic borosilicate glass microfiber with 96% void volume	Cellulose paper that collapses when wet with oil/water
Housing Coating	Alocrom anti-corrosion passivation + electrostatic epoxy powder	Uncoated cast aluminum prone to internal oxidation
Pressure Drop	Initial dry ΔP <0.07 bar; aerodynamic smooth internal air channels	Rough internal cast edges causing high friction and >0.2 bar ΔP
Element Sealing	Dual O-ring push-fit seal prevents any bypass leakage	Single flat gasket prone to bypass under vibration

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- 100% Hydrostatic pressure test of cast housings at 24 bar g (1.5× working pressure)
- ISO 12500-1 certified aerosol challenge oil penetration testing on filter elements
- Pressure drop differential verification on calibrated air flow test bench
- Differential pressure indicator calibration and pop-up trip point test

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 10 Years warranty on aluminum filter housings; 12 months on differential indicator. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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